

Formal Design Specification: Capability Embedding Framework

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1 System Overview

This document specifies the formal design of the capability composition model. State transitions are governed by dense vector mappings.

2 Vector Space Model

Let d be the dimensionality of the state space. Each capability C is embedded in a vector space $\mathbb{R}^{2d+3}$:

$$\vec{v}_C = \begin{bmatrix} \vec{p} \\ \vec{e} \\ c \\ r \\ a \end{bmatrix} \in \mathbb{R}^{2d+3} \quad (1)$$

where $\vec{p} \in \mathbb{R}^d$ represents preconditions, $\vec{e} \in \mathbb{R}^d$ represents effects, $c \in \mathbb{R}^+$ denotes cost, $r \in [0, 1]$ denotes reliability, and $a \in [0, 1]$ denotes availability.

3 Composition Operators

Given C_1 and C_2 , the composition $C_{12} = C_2 \circ C_1$ is defined component-wise as:

$$p_{12,i} = \max(p_{1,i}, p_{2,i} - e_{1,i}) \quad \forall i \in \{1, \dots, d\} \quad (2)$$

$$e_{12,i} = \max(e_{1,i}, e_{2,i}) \quad \forall i \in \{1, \dots, d\} \quad (3)$$

$$c_{12} = c_1 + c_2 \quad (4)$$

$$r_{12} = r_1 \times r_2 \quad (5)$$

$$a_{12} = \min(a_1, a_2) \quad (6)$$